

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial 11 – exam rehearsal - Solutions

Question 1

A factory performs continuous fermentation to produce a certain product, in a series of 3 fermenters each of a different volume, that are connected serially one after the other.

The volume of the first fermenter is 500 liter and it receives fresh feed flow at rate of 200 liter per hour, which contains carbon source at concentration of 10%. The flow exiting this fermenter is streamed into the second fermenter, that its volume is 700 liters. The flow exiting the second fermenter is streamed into the third fermenter, that its volume is 1200 liters. The liquid exiting the third fermenter contains 2 ppm of substrate, and continuously streamed to separation processes which aims to isolate and separate the product. The total yield of the product separation processes is 85.3%.

The biomass yield on this carbon source is $0.5 \frac{gX}{gS}$, product yield by the biomass is $0.45 \frac{gP}{gX}$, and the biomass yield on oxygen is $1.6 \frac{gX}{gO_2}$. The maximal growth rate of the microorganisms in this system is 0.5 per hour and Monod constant equals to $700 \frac{mg}{lit}$. However, in this fermentation system, the carbon source is the limiting factor only in the first and third fermenters, and not in the second fermenter where the limiting factor is other.

The fermenters receive regular air at pressure of 2 atmospheres, while in case of malfunction in the aeration system – until the regular air supply return to normal, the fermenters are supplied with oxygen enriched air from backup gas tank, at pressure of 1.6 atmospheres and oxygen concentration of 40%. It can be assumed that the oxygen transfer constants are not changed as a result of the different oxygen concentrations in the air supplied into the fermenters.

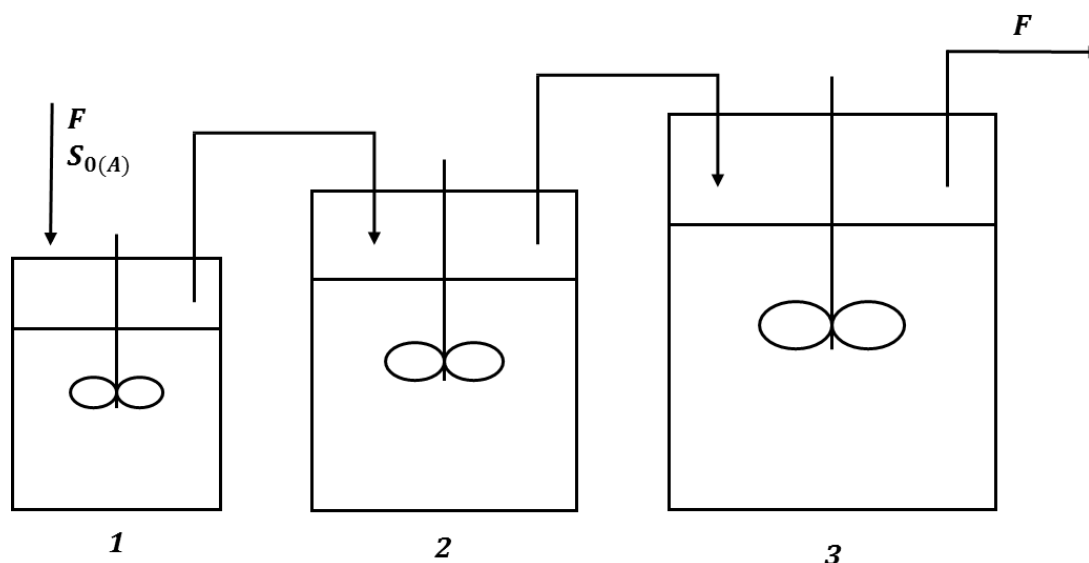
From a look on the computer records of the oxygen electrode readings in the third fermenter, it seems that there was a short aeration stop during the fermenter operation. At the moment

when the regular aeration paused, the oxygen dissolved concentration in the fermenter was 50%, and after 6 seconds the electrode reading was reduced to 10% (this reading is still higher than the critical oxygen concentration in this system, so growth didn't stop). Once the electrode reading reached 10%, the enriched air supply from the gas tanks came into action and the dissolved oxygen concentration in the fermenter increased gradually until a value of C_E , and stabilized on this concentration. After the regular air supply was returned, the electrode reading returned gradually to its original reading of 50% and stabilized on this reading.

It can be assumed that the death rate in the system is negligible, that the substrate consumption for maintenance is negligible, that the product decomposition is negligible, and that the product is mainly nitrogen-based.

- A. What is the dissolved oxygen saturation concentration in the third fermenter?**
- B. What was the electrode reading when the dissolved oxygen concentration in the third fermenter was stabilized on the value of C_E ?**
- C. What is the growth rate of the microorganisms in the second fermenter?**
- D. The factory works 300 days a year and aims to produce at least 30 tons of separated pure product per year. Can this fermentation system meet this demand?**

Solution:



Fermentation

$$V_1 = 500 \text{ lit}$$

$$F = 200 \frac{\text{lit}}{\text{hr}}$$

$$S_0 = 100 \frac{\text{gS}}{\text{lit}}$$

$$V_2 = 700 \text{ lit}$$

$$\eta = 85.3\% = 0.853$$

$$Y_{\frac{X}{S}} = 0.5 \frac{\text{gX}}{\text{gS}}$$

$$Y_{\frac{P}{X}} = 0.45 \frac{\text{gP}}{\text{gX}}$$

$$Y_{\frac{X}{O}} = 1.6 \frac{\text{gX}}{\text{gO}_2}$$

$$\mu_{max} = 0.5 \frac{1}{\text{hr}}$$

$$K_S = 700 \frac{\text{mg}}{\text{lit}} = 0.7 \frac{\text{g}}{\text{lit}}$$

Air

$$\%O_2 = 21\% = 0.21$$

$$P = 2 \text{ atm}$$

Air during malfunction

$$\%O_2 = 40\% = 0.4$$

$$P = 1.6 \text{ atm}$$

Fermenter 3

$$V_3 = 1200 \text{ lit}$$

$$C_{L_3} = 0.5 \cdot C_3^*$$

$$C'_{L_3}(t = 6 \text{ sec}) = 0.1 \cdot C_3^*$$

$$S_{F_3} = 2 \text{ ppm} = 0.002 \frac{\text{gS}}{\text{lit}}$$

A. $C_3^* = ?$

During steady state:

$$K_L a_3 \cdot (C_3^* - C_{L_3}) = \frac{\mu_3 \cdot X_{F_3}}{Y_{\frac{X}{O}}}$$

Moreover, during the aeration stop:

$$\frac{\Delta C_{L_3}}{\Delta t} = -OUR$$

C_L is given as related to C^* so we will be able to find C^* if we will find OUR .

$$OUR_3 = \frac{\mu_3 \cdot X_{F_3}}{Y_{\frac{X}{O}}}$$

Finding μ_3 and X_{F_3} :

In fermenter 3 we already discovered (according to equations which we developed in tutorial 9, question 1) that:

$$X_{F_3} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_3}) = 0.5 \frac{gX}{gS} \cdot (100 - 0.002) \frac{gS}{lit} = 50 \frac{gX}{lit}$$

From Monod equation:

$$\mu_3 = 0.5 \frac{1}{hr} \cdot \left(\frac{0.002 \frac{gS}{lit}}{0.7 \frac{gS}{lit} + 0.002 \frac{gS}{lit}} \right) = 0.0014 \frac{1}{hr}$$

Place number in the equation ($\frac{\Delta C_{L_3}}{\Delta t} = -OUR$) to find C_3^* :

$$\frac{(0.1 - 0.5) \cdot C_3^*}{6 \text{ sec} \cdot \frac{1 \text{ hr}}{3600 \text{ sec}}} = - \frac{0.0014 \frac{1}{hr} \cdot 50 \frac{gX}{lit}}{1.6 \frac{gX}{gO_2}}$$

$$C_3^* = 1.82 \cdot 10^{-4} \frac{gO_2}{lit}$$

B. $\frac{C_E}{C_3^*} \cdot 100 = ?$

The electrode reading does not return back to the original reading (50%) before the regular air pause, since the change in the characteristics of the supplied enriched air affect the value of C^* (although when steady state will re-stabilized - both OUR and OTR will be the same as before the regular air pause).

We will assume that all the stabilization process is very short, thus OUR does not change, hence:

$$OTR_{new}^* = OTR \text{ (* meaning in the enriched air.)}$$

$$K_L a \cdot (C_{new}^* - C_E)_{new} = K_L a \cdot (C_3^* - C_{L_3})$$

$K_L a$ does not change, therefore it can be reduced to:

$$(C_{new}^* - C_E) = (C_3^* - C_{L_3}) = 0.5 \cdot C_3^* \rightarrow C_E = C_{new}^* - 0.5 \cdot C_3^*$$

Find C_{new}^* from Henry's law:

$$C_{new}^* = \frac{P_{new} \cdot X_{O_2_{new}}}{H}$$

To find Henry's constant - we will use the initial conditions:

$$H = \frac{P_{regular} \cdot X_{O_2 regular}}{C_3^*} = \frac{2 \text{ atm} \cdot 0.21}{1.82 \cdot 10^{-4} \frac{gO_2}{lit}} = 2307 \frac{lit \cdot atm}{gO_2}$$

$$C_{new}^* = \frac{1.6 \text{ atm} \cdot 0.4}{2307 \frac{lit \cdot atm}{gO_2}} = 2.77 \cdot 10^{-4} \frac{gO_2}{lit}$$

$$C_E = C_{new}^* - 0.5 \cdot C_3^*$$

The electrode was not recalibrated, so the normalization is by the original C_3^* :

$$\frac{C_E}{C_3^*} = \frac{(C_{new}^* - 0.5 \cdot C_3^*)}{C_3^*} = \left(\frac{C_{new}^*}{C_3^*} - 0.5 \right) = \left(\frac{2.77 \cdot 10^{-4} \frac{gO_2}{lit}}{1.82 \cdot 10^{-4} \frac{gO_2}{lit}} - 0.5 \right) = 1.022$$

$$\frac{C_E}{C_3^*} \cdot 100 = 1.022 \cdot 100 = \mathbf{102.2\%}$$

The electrode will show reading of 102.2%.

C. $\mu_2 = ?$

In fermenter 1 steady state equations are:

$$\mu_1 = D_1$$

$$X_{F_1} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_1})$$

$$P_{F_1} = Y_{\frac{P}{X}} \cdot X_{F_1} = Y_{\frac{P}{S}} \cdot (S_0 - S_{F_1})$$

In fermenter 2 steady state equations are:

$$\mu_2 = D_2 \cdot \left(1 - \frac{X_{F_1}}{X_{F_2}} \right)$$

$$X_{F_2} = Y_{\frac{X}{S}} \cdot (S_0 - S_{F_2})$$

$$P_{F_2} = Y_P \cdot X_{F_2} = Y_P \cdot (S_0 - S_{F_2})$$

We want to find X_{F_1} and X_{F_2} :

$$\mu_1 = D_1 = \frac{F}{V_1} = \frac{200 \frac{lit}{hr}}{500 lit} = 0.4 \frac{1}{hr}$$

According to Monod equation:

$$S_{F_1} = \frac{\mu_1 \cdot K_S}{(\mu_{max} - \mu_1)} = \frac{0.4 \frac{1}{hr} \cdot 0.7 \frac{g}{lit}}{(0.5 - 0.4) \frac{1}{hr}} = 2.8 \frac{gS}{lit}$$

$$X_{F_1} = 0.5 \frac{gX}{gS} \cdot (100 - 2.8) \frac{gS}{lit} = 48.6 \frac{gX}{lit}$$

We can't find S_{F_2} at the moment since we don't know μ_2 , therefore, we will focus on fermenter 3:

$$\mu_3 = D_3 \cdot \left(1 - \frac{X_{F_2}}{X_{F_3}}\right) = \left(\frac{F}{V_3}\right) \cdot \left(1 - \frac{X_{F_2}}{X_{F_3}}\right)$$

$$0.0014 \frac{1}{hr} = \left(\frac{200 \frac{lit}{hr}}{1200 lit}\right) \cdot \left(1 - \frac{X_{F_2}}{50 \frac{gX}{lit}}\right)$$

$$X_{F_2} = 49.58 \frac{gX}{lit}$$

If we'll go back to the equation of growth rate in the second fermenter, and place the found data:

$$\mu_2 = D_2 \cdot \left(1 - \frac{X_{F_1}}{X_{F_2}}\right) = \left(\frac{F}{V_2}\right) \cdot \left(1 - \frac{X_{F_1}}{X_{F_2}}\right) = \left(\frac{200 \frac{lit}{hr}}{700 lit}\right) \cdot \left(1 - \frac{48.6 \frac{gX}{lit}}{49.58 \frac{gX}{lit}}\right)$$

$$\mu_2 = 0.0056 \frac{1}{hr}$$

D. $G_{P_{clean}} = ?$

$$G_P = F \cdot P_{F_3} = F \cdot Y_{\frac{P}{X}} \cdot X_{F_3} = 200 \frac{\text{lit}}{\text{hr}} \cdot 0.45 \frac{\text{gP}}{\text{gX}} \cdot 50 \frac{\text{gX}}{\text{lit}} = 4500 \frac{\text{gP}}{\text{hr}}$$

$$G_{P_{clean}} = 4500 \frac{\text{gP}}{\text{hr}} \cdot 0.853 = 3838.5 \frac{\text{gP}}{\text{hr}} \cdot 24 \frac{\text{hr}}{\text{day}} \cdot 300 \frac{\text{day}}{\text{year}} = 27.6 \frac{\text{ton}}{\text{year}} < 30 \frac{\text{ton}}{\text{year}}$$

That means – the system can't meet the demand.

Question 2

A factory grows yeast in 10,000 liters batch fermenter. After medium preparation and before its sterilization, the medium contains 10^5 spores per liter, while the sterilization demands are that the contamination probability will be 1 to 1000 fermentations.

The sterilization of the medium is carried continuously during the filling up of the fermenter, as the medium flows through a continuous sterilizer before entering and filling the fermenter (it can be assumed that the time it takes for the medium in the sterilizer to reach the sterilization temperature is very quick). The volume of the continuous sterilizer is 1,125 liters. The time it takes to fill the fermenter as described, is 80 minutes.

From professional literature you found out that sterilization of contaminating spores at temperature of 127°C for 11 minutes allows sterilization level of $1.85 \cdot 10^{-6}$, and that sterilization of these spores at temperature of 145°C for 12 minutes allows sterilization level of $3.2 \cdot 10^{-9}$.

The correlation between the sterilization constant (K_d) to the sterilization temperature is described by Arrhenius equation, meaning:

$$K_d = A \cdot e^{-E_a/RT}$$

A – empiric constant (min^{-1})

T – sterilization temperature ($^\circ\text{K}$)

E_a – activation energy ($\frac{\text{cal}}{\text{mole}}$)

R – universal gas constant ($1.98 \frac{\text{cal}}{\text{mole} \cdot ^\circ\text{K}}$)

What is the sterilization temperature to comply with the demand?

Solution:

Given:

<u>Literature</u>		<u>Fermentation</u>
$T = 127^{\circ}\text{C} = 400^{\circ}\text{K}$	$T = 145^{\circ}\text{C} = 418^{\circ}\text{K}$	$V_{ferm} = 10,000 \text{ L}$
$t_{st127^{\circ}\text{C}} = 11 \text{ min}$	$t_{st145^{\circ}\text{C}} = 12 \text{ min}$	$N'_0 = 10^5 \frac{\text{Spore}}{\text{L}}$
$\frac{N_f}{N_0} = 1.85 \cdot 10^{-6}$	$\frac{N_f}{N_0} = 3.2 \cdot 10^{-9}$	$N_f = \frac{1 \text{ Spore}}{1000 \text{ Ferm}}$
		$V_{sterilizer} = 1,125 \text{ L}$
		$t_{fill} = 80 \frac{\text{min}}{\text{Ferm}}$

 $T_{st} = ?$

Solution strategy:

- The sterilization temperature will be extracted from Arrhenius equation. We need to find K_d , A and Ea.
- Find the sterilization constant (K_d) in the given fermentation using the sterilization equation.
 - N_f is given in the question.
 - N_0 can be calculated from the fermenter volume.
 - Calculate the staying time from the medium flow rate and the sterilizer volume.
 - Calculate the medium flow rate from the medium volume and the time it takes to fill the fermenter.
- To find A and Ea we need the sterilization constants of the given temperature from the literature.
 - We will calculate the constants by the sterilization equation according to the sterilization period and the sterilization level described in the literature.
 - Write the two equations with two unknowns, isolate Ea first (by dividing the equations) and place to find A.
- Place K_d of the fermentation with the constant we found from the literature data.

Calculate the sterilization temperature from Arrhenius equation:

$$K_d = A \times e^{\frac{-E_a}{RT}}$$

First, calculate the sterilization constant that is required in the process described in the question, from the killing equation:

$$\ln\left(\frac{N_f}{N_0}\right) = -k_d \times t_{st}$$

N_f is given in the question. We will calculate N_0 according to N'_0 :

$$N_0 = N'_0 \times V_{ferm}$$

$$N_0 = 10^5 \frac{spore}{lit} \times 10,000 \frac{lit}{ferm} = 10^9 \frac{spore}{Ferm}$$

The staying time can be found from the medium flow rate through the sterilizer:

$$t_{st} = \frac{V_{sterilizer}}{Q}$$

While the medium flow rate can be found from the fermenter volume and the time it takes to fill it:

$$Q = \frac{V_{ferm}}{t_{fill}} = \frac{10,000 \frac{lit}{ferm}}{80 \frac{min}{ferm}} = 125 \frac{lit}{min}$$

Therefore, the staying time is:

$$t_{st} = \frac{V_{sterilizer}}{Q} = \frac{1125 \text{ lit}}{125 \frac{lit}{min}} = 9 \text{ min}$$

Hence, the sterilization constant in this fermentation is:

$$\ln\left(\frac{N_f}{N_0}\right) = -k_d \times t_{st}$$

$$\ln\left(\frac{0.001 \frac{spore}{ferm}}{10^9 \frac{spore}{ferm}}\right) = -k_d \times 9 \text{ min} \quad \rightarrow \quad k_d = 3.07 \frac{1}{min}$$

To find Ea and A - use literature data. First we will find the sterilization constant in each temperature:

$$k_d = \frac{\ln\left(\frac{N_f}{N_0}\right)}{-t_{st}}$$

$$k_{d_{418^\circ K}} = \frac{\ln(3.2 \cdot 10^{-9})}{-12 \text{ min}} = 1.63 \frac{1}{min}$$

$$k_{d_{400^\circ K}} = \frac{\ln(1.85 \cdot 10^{-6})}{-11 \text{ min}} = 1.2 \frac{1}{min}$$

Back to Arrhenius equation:

$$K_d = A \cdot e^{\frac{-Ea}{RT}}$$

$$1.63 \frac{1}{min} = A \cdot e^{\left(\frac{\frac{-Ea}{cal}}{1.98 \frac{cal}{mole \cdot K} \cdot 418^\circ K}\right)}$$

$$1.2 \frac{1}{min} = A \cdot e^{\left(\frac{\frac{-Ea}{cal}}{1.98 \frac{cal}{mole \cdot K} \cdot 400^\circ K}\right)}$$

Divide the equations to find Ea:

$$\frac{1.63 \frac{1}{min}}{1.2 \frac{1}{min}} = \frac{e^{\frac{\frac{-Ea}{cal}}{827.64 \frac{cal}{mole}}}}{e^{\frac{\frac{-Ea}{cal}}{792 \frac{cal}{mole}}}}$$

$$Ea = 5628.2 \frac{cal}{mole}$$

To find A, place Ea in one of the equations:

$$1.2 \frac{1}{\min} = A \cdot e^{\left(\frac{-5628.2 \frac{\text{cal}}{\text{mole}}}{1.98 \frac{\text{cal}}{\text{mole} \cdot \text{K}} \cdot 400^\circ \text{K}} \right)}$$

$$A = 1463.56 \frac{1}{\min}$$

Now, use Arrhenius equation with the killing constant required for the process described in the question, to find the sterilization temperature:

$$K_d = A \cdot e^{\frac{-E_a}{RT}}$$

$$3.07 \frac{1}{\min} = 1463.56 \frac{1}{\min} \cdot e^{\left(\frac{-5628.2 \frac{\text{cal}}{\text{mole}}}{1.98 \frac{\text{cal}}{\text{mole} \cdot \text{K}} \cdot T_{st}} \right)}$$

$$T_{st} = 460.93^\circ \text{K} \cong 461^\circ \text{K}$$

$$T_{st} = 188^\circ \text{C}$$

The sterilization should be carried in temperature of 188 degrees Celsius.

Question 3

A factory performs batch fermentation of yeast for the production of extracellular protein in 900-liter fermenter. The carbon source in the fermentation is glycerol and its initial concentration is 2%. The yeast is inoculated in initial concentration of 400 ppm and after 2 hours of lag, the yeast starts logarithmic growth. The yeast doubling time during logarithmic stage is 1.5 hour. At the end of the logarithmic growth, the fermenter is harvested, and its contents transferred to separation processes to produce the protein in a total yield of 86.9%. After discharging the fermenter, a process of cleaning, filling and sterilization is performed, so that when finished - the fermenter is ready for the next fermentation which immediately inoculated. The factory produces 168 kg of clean protein per month, while one working month includes 30 days and the factory works 24 hours per day.

Regular air is supplied into the fermenter at flow rate of 1500 LPM, temperature of 35°C and pressure of 1 atmosphere. It can be assumed that the air acts as ideal gas and that the air flow entering and exiting the fermenter are at the same rate.

Additional data: the oxygen solubility percentage at the end of the fermentation is 13%, the specific product production rate is $0.24 \frac{gP}{gX \cdot hr}$, the general biomass production rate is 1150 gram biomass per hour, the volumetric substrate consumption rate is $2.95 \frac{gS}{lit \cdot hr}$ and the glycerol is completely consumed during fermentation.

- A. What is the time period for cleaning and preparation in each fermentation cycle?**
B. What is the product yield on oxygen?

Solution:

Given:

<u>Fermentation</u>	<u>Yeast</u>	<u>Air</u>
$P_0 = 0 \frac{gP}{L}$	$X_0 = 400ppm = 0.4 \frac{gX}{L}$	$Q_{air} = 1500 \frac{L_{air}}{min}$
$S_0 = 2\% = 20 \frac{gS}{L}$	$t_d = 1.5hr$	$T = 35^\circ C = 308^\circ K$
$V = 900 L$	$q_P = 0.24 \frac{gP}{gX \times hr}$	$P = 1atm$
$\Delta t_{lag} = 2hr$	$R_X = 1150 \frac{gX}{hr}$	$\% Solubility_f = 13\%$
$\frac{W_{P_{clean}}}{month} = 168 \frac{KgP_{clean}}{month}$	$\frac{R_S}{V} = 2.95 \frac{gS}{L \times hr}$	
$\eta = 86.9\% = 0.869 \frac{KgP_{clean}}{KgP}$		
$S_f = 0 \frac{gS}{L}$		

- A. $t_{maintenance} = ?$**

Solution steps:

1. The maintenance time can be extracted from the length of each fermentation cycle. For that we need to find the duration of one fermentation cycle and that of the logarithmic growth.
2. The time period of one fermentation cycle can be calculated from the monthly output. For that we need to find the amount of product obtained in a single fermentation cycle.
3. The amount of product obtained in one fermentation cycle can be calculated from the production yield. For that we need to find the production yield and the biomass concentration at the end of the growth.
4. The biomass concentration at the end of the growth can be found from the growth yield.
5. The growth yield can be found from the general biomass production rate and the volumetric substrate consumption rate.
6. The production yield can be extracted from the specific protein production rate. For that we need to growth rate.
7. The growth rate can be found from the generation time.

The maintenance time is part of the fermentation cycle:

$$t_T = \Delta t_{lag} + \Delta t_{log} + \Delta t_{maintenance} \rightarrow \Delta t_{maintenance} = t_T - (\Delta t_{lag} + \Delta t_{log})$$

The protein production yield can be calculated by the specific production rate given in the question:

$$q_P = \mu \cdot Y_{\frac{P}{X}}$$

We need to growth rate – calculate it from the doubling time of the yeast in the fermentation:

$$\mu = \frac{\ln 2}{t_d}$$

$$\mu = \frac{\ln 2}{1.5 \text{ hr}} = 0.462 \frac{1}{\text{hr}}$$

Place the growth rate to find the protein production yield:

$$Y_{\frac{P}{X}} = \frac{q_P}{\mu}$$

$$Y_{\frac{P}{X}} = \frac{0.24 \frac{gP}{gX \cdot hr}}{0.462 \frac{1}{hr}} = 0.52 \frac{gP}{gX}$$

Find ΔP from the production yield:

$$Y_{\frac{P}{X}} = \frac{\Delta P}{\Delta X} = \frac{\Delta P}{X_f - X_0}$$

The yeast concentration at the beginning of the growth is given in the question. The yeast concentration at the end of the growth we can calculate from the growth yield:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} = \frac{X_f - X_0}{S_0 - S_f}$$

The growth yield can be calculated from the general biomass production rate and the volumetric substrate consumption rate, given in the question:

$$Y_{\frac{X}{S}} = \frac{R_X}{\frac{R_S}{V} \cdot V}$$

$$Y_{\frac{X}{S}} = \frac{1150 \frac{gX}{hr}}{2.95 \frac{gS}{lit \cdot hr} \cdot 900 lit} = 0.43 \frac{gX}{gS}$$

Hence, the final biomass concentration is:

$$0.43 \frac{gX}{gS} = \frac{(X_f - 0.4) \frac{gX}{lit}}{(20 - 0) \frac{gS}{lit}}$$

$$X_f = 9 \frac{gX}{lit}$$

Place the yeast concentration at the end of the fermentation in the production yield:

$$\Delta P = Y_{\frac{P}{X}} \cdot (X_f - X_0)$$

$$\Delta P = 0.52 \frac{gP}{gX} \cdot (9 - 0.4) \frac{gX}{lit} = 4.47 \frac{gP}{lit}$$

It is given that the protein is extracellular, therefore:

$$\Delta P = (P_f - P_0) \rightarrow P_f = \Delta P + P_0$$

$$P_f = (4.47 + 0) \frac{gP}{lit} = 4.47 \frac{gP}{lit}$$

By multiplying the final concentration with the fermenter volume, we can find the protein amount that is produced in one fermentation:

$$\frac{W_{P_f}}{ferm} = P_f \cdot V$$

$$\frac{W_{P_f}}{ferm} = \frac{4.47 \frac{gP}{lit} \cdot 900 lit}{1000 \frac{KgP}{KgP}} = 4 \frac{KgP}{ferm}$$

This amount of protein is generated in one fermentation cycle, so by dividing the monthly protein amount in this amount - we can find the numbers of fermentation cycles done in a month, hence we can calculate the time period of one fermentation cycle.

Convert the amount of clean protein obtained in one month to the amount of protein produced per month:

$$\frac{W_P}{month} = \frac{\left(\frac{W_{P_{clean}}}{month} \right)}{\eta}$$

$$\frac{W_P}{month} = \frac{168 \frac{KgP_{clean}}{month}}{0.869 \frac{KgP_{clean}}{KgP}} = 193 \frac{KgP}{month}$$

From the monthly output of clean protein, calculate the amount of protein produces per month:

$$t_T = \frac{\frac{W_{P_f}}{ferm} \cdot \frac{hr}{day} \cdot \frac{day}{month}}{\frac{W_P}{month}}$$

$$t_T = \frac{4 \frac{KgP}{ferm} \cdot 24 \frac{hr}{day} \cdot 30 \frac{day}{month}}{193 \frac{KgP}{month}} = 15 \frac{hr}{ferm}$$

Use the logarithmic growth equation to find the period of logarithmic growth:

$$\ln\left(\frac{X_f}{X_0}\right) = \mu \cdot \Delta t_{log}$$

$$\Delta t_{log} = \frac{\ln\left(\frac{9 \frac{gX}{L}}{0.4 \frac{gX}{L}}\right)}{0.462 \frac{1}{hr}} = 6.7 \text{ hr}$$

Therefore, the maintenance time:

$$\Delta t_{maintenance} = t_T - (\Delta t_{lag} + \Delta t_{log})$$

$$\Delta t_{maintenance} = (15 - (2 + 6.7))hr$$

$$\Delta t_{maintenance} = 6.3hr$$

B. $Y_{\frac{P}{O}}$ = ?

Solution steps:

1. Calculate the product yield on oxygen by multiplication of the biomass yield on oxygen ($Y_{\frac{X}{O}}$) with the product yield ($Y_{\frac{P}{X}}$) that was calculated in section A.
2. The biomass yield on oxygen can be extracted from the maximal oxygen uptake rate.
3. Calculated the maximal oxygen uptake rate from the air flow rate and the oxygen solubility percentage at the end of the fermentation.

The biomass yield on oxygen can be calculate from the oxygen uptake rate:

$$O.U.R = \frac{\mu \cdot X}{Y_{\frac{X}{O}}}$$

The growth rate and the maximal biomass concentration (X_f) were calculated in section A.

The oxygen uptake rate we can calculate from the oxygen solubility:

$$O.U.R = \left(\frac{Q_{O_2}}{V} \right)_{in} \cdot (\%Solubility)$$

The question gives the solubility percentage at the end of the fermentation. In order to find the oxygen uptake rate, we need to convert the air flow rate to volumetric oxygen flow rate:

$$\begin{aligned} \left(\frac{Q_{O_2}}{V} \right)_{in} &= \frac{Q_{air}}{V} \cdot \%O_2 \cdot \frac{P}{T \cdot R} \cdot Mw_{O_2} \\ \left(\frac{Q_{O_2}}{V} \right)_{in} &= \frac{1500 \frac{lit_{air}}{min}}{900 lit} \cdot 60 \frac{min}{hr} \cdot 0.21 \frac{lit_{O_2}}{lit_{air}} \cdot \frac{1 atm}{308^\circ K \cdot 0.082 \frac{lit_{O_2} \cdot atm}{K \cdot mole_{O_2}}} \cdot 32 \frac{g_{O_2}}{mole_{O_2}} \\ \left(\frac{Q_{O_2}}{V} \right)_{in} &= 26.61 \frac{g_{O_2}}{lit \cdot hr} \end{aligned}$$

It is given that the solubility percentage at the end of the fermentation is 13%. This given solubility is the minimal required to enable enough oxygen to the biomass at the end of this specific fermentation. therefore, we can use it to calculate the oxygen uptake of the yeast at the end of the fermentation:

$$\begin{aligned} O.U.R_f &= \left(\frac{Q_{O_2}}{V} \right)_{in} \cdot (\%Solubility)_f \\ O.U.R_f &= 26.61 \frac{g_{O_2}}{lit \cdot hr} \cdot \frac{13}{100} = 3.46 \frac{g_{O_2}}{lit \cdot hr} \end{aligned}$$

The growth rate and the yeast concentration at the end of the fermentation were calculated in section A. Therefore, the biomass yield on oxygen:

$$O.U.R_f = \frac{\mu \cdot X_f}{Y_{X/O}}$$

$$Y_{\frac{X}{O}} = \frac{0.462 \frac{1}{hr} \cdot 9 \frac{gX}{lit}}{3.46 \frac{gO_2}{lit \cdot hr}} = 1.2 \frac{gX}{gO_2}$$

Hence, the product yield on the oxygen is:

$$Y_{\frac{P}{O}} = Y_{\frac{P}{X}} \times Y_{\frac{X}{O}} = 0.52 \frac{gP}{gX} \times 1.2 \frac{gX}{gO_2}$$

$$Y_{\frac{P}{O}} = 0.624 \frac{gP}{gO_2}$$